



AI Perimeter Intrusion Detection Network

IoT EL

Go, change the world

EL Phase-I

Presentation Outline

- Introduction
- Literature Review
- Problem Definition
- Objectives
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- Proposed Outcomes
- Conclusion
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Introduction

What is Perimeter Intrusion Detection?

The process of sensing, identifying, and responding to unauthorized entry attempts across boundaries like factories, farms, or military outposts.

The Scale of the Problem

- **₹8,400 crore loss** annually in India due to theft and vandalism.
- **60%+ PIR systems** disabled within 2 weeks due to false alerts.
- **High Cost:** Existing systems cost ₹40k–₹2L per node.
- **78% of IoT deployments** lack classification intelligence.

Why This Project Matters

A 3–4 node ESP32 mesh system fusing PIR, vibration, and ultrasonic sensors.

- **LLM Classification:** Identifies human, animal, or vehicle.
- **Low Cost:** Under ₹2,800 total hardware cost.
- **Offline Capability:** Works without internet; Telegram alerts when online.

Why ESP-NOW?

Connectionless 2.4GHz peer-to-peer protocol ideal for router-less environments.

- **Sub-5ms latency** vs 100-300ms for Wi-Fi.
- **200m range** and zero infrastructure dependency.

Literature Review

Topic	Paper	Year	Summary	Key Gap
IoT-based intrusion detection using sensor fusion	R. Kumar et al., <i>"Multi-Sensor Fusion for Perimeter Intrusion Detection in IoT Networks,"</i> IEEE Internet of Things Journal, vol. 10, no.4	2023	• Proposes a fusion of PIR and acoustic sensors on ESP8266 nodes. Achieved 87% classification accuracy for human vs. non-human using threshold logic.	• No mesh networking; single-node architecture; no AI/LLM-based classification.
ESP-NOW mesh for low-latency IoT	S. Zhao and L. Wei, <i>"Low-Latency Peer-to-Peer Communication for Edge IoT Using ESP-NOW Protocol,"</i> IEEE Access, vol. 11	2023	• Demonstrated sub-5ms latency in ESP-NOW mesh topologies of up to 6 nodes . Measured packet loss under 0.3% in open-field tests.	• Study limited to communication benchmarking; no sensor payload or security application explored.

Literature Review

Topic	Paper	Year	Summary	Key Gap
LLM agents for autonomous IoT decision-making	A. Gupta et al., " <i>Towards LLM-Augmented Edge Intelligence for Autonomous IoT Systems</i> ," Proceedings of IEEE ISCAS	2024	• Demonstrated GPT-4 acting as a reasoning agent for sensor data classification in smart home contexts. Accuracy improved 22% over rule-based systems.	• Applied only to indoor environments; no outdoor perimeter or multi-node sensor fusion evaluated.
Vibration-based intrusion classification	P. Nair and T. Krishnan, " <i>Ground Vibration Signature Analysis for Human-Vehicle Discrimination</i> ," IEEE Sensors Journal, vol. 23, no. 8	2023	• FFT analysis of SW-420 vibration sensor signals distinguished human footsteps (8–20 Hz) from vehicle (1–5 Hz) with 91% accuracy in controlled conditions.	• Standalone vibration sensor only; no fusion with PIR or ultrasonic; no wireless transmission.

Literature Review

Topic	Paper	Year	Summary	Key Gap
Ultrasonic ranging for object size estimation	M. El-Shafei et al., <i>"Object Size Estimation Using HC-SR04 for Embedded Classification Systems,"</i> IEEE Embedded Systems Letters, vol. 15	2024	• Used time-of-flight variance from HC-SR04 to estimate object width as a supplementary classification feature. Improved discrimination between small animals and humans by 17%	• Only tested in lab corridors; no outdoor field validation; not integrated with PIR or wireless mesh.

Key Understandings

Literature Insights: What the Field Tells Us vs. Unsolved Gaps

Insight 1: Modality Fusion

Single sensors (PIR/Vibration/Ultrasonic) only detect; they cannot classify. Reliable identification requires fusion of ≥ 3 modalities.

Insight 2: LLM over Rule-Engines

LLMs achieve 91.3% accuracy by reasoning contextually over ambiguous sensor data, significantly outperforming static rule-based thresholds.

Insight 3: Edge Computing

Edge-first architectures reduce latency by 82%. Processing at the gateway ensures immediate response regardless of internet availability.

Insight 4: Autonomous Orchestration

LLM agents autonomously decide sensor activation and event classification, enabling predictive, direction-aware perimeter security.

Insight 5: Hardware Gap (The Novelty)

No existing work utilizes low-cost ESP32-class microcontrollers (₹350) for LLM-integrated mesh systems. Most rely on expensive Raspberry Pi/GPU servers.

Our Project: Filling the Technical Gap

Combining three-sensor fusion, ESP-NOW mesh, and autonomous LLM classification on ultra-low-cost nodes (under ₹2,800/node).

Problem Definition

Problem Statement

To design and validate a cost-effective, offline-capable, multi-node perimeter intrusion detection system that moves beyond binary motion sensing to achieve intelligent classification (human, animal, or vehicle) using an LLM agent, enabling autonomous alarm management at under ₹2,800 per node.

Context of the Problem

- Existing perimeter security systems generate identical alerts for harmless movements and actual intrusions, making accurate threat identification difficult.
- High false alarm rates in PIR-based systems lead to alarm fatigue, causing security personnel to ignore or delay responses to real threats.
- Most modern surveillance systems rely on continuous internet connectivity, making them ineffective in remote or low-network areas where security is critically needed.

Significance of Addressing This

- Prevents **operational uselessness** from false-alarm fatigue.
- Enables **triage & proportional response** via intruder classification.
- Eliminates coverage gaps with **mesh redundancy**.
- **Offline capability** ensures resilience during outages or jamming.

Objectives

Main Objective

To design, implement, and validate a 3–4 node ESP32 ESP-NOW mesh perimeter intrusion detection network that autonomously classifies intruders (human, animal, vehicle) using fused sensor data and LLM agents, targeting >85% accuracy and <4s latency.

Specific Objectives

- Deploy 3–4 nodes in ESP-NOW mesh with <100ms latency and 15m+ range.
- Instrument nodes with PIR, vibration, and ultrasonic sensors.
- Design LLM prompt architecture for JSON multi-node sensor payloads.
- Implement dual-mode alerting: (a) Local offline buzzer/LED; (b) Cloud Telegram Bot.
- Evaluate with 60-event test set (human, animal, vehicle) reporting per-class metrics.

Expected Contributions

- Replicable reference architecture for sub-₹3,000 intelligent nodes.
- LLM agent integration with ESP32 for real-time edge classification.
- Novel labelled sensor signature dataset for perimeter security events.

Methodology

Design Approach

Mixed-methods experimental design combining empirical hardware calibration (sensor/mesh latency) with quantitative evaluation of classification performance against labelled intrusion datasets.

Data Collection Methods

Nodes collect (1) PIR trigger pulse, (2) vibration frequency, and (3) ultrasonic distance. Time-stamped data is transmitted via ESP-NOW to a gateway for JSON aggregation.

Data Analysis Techniques

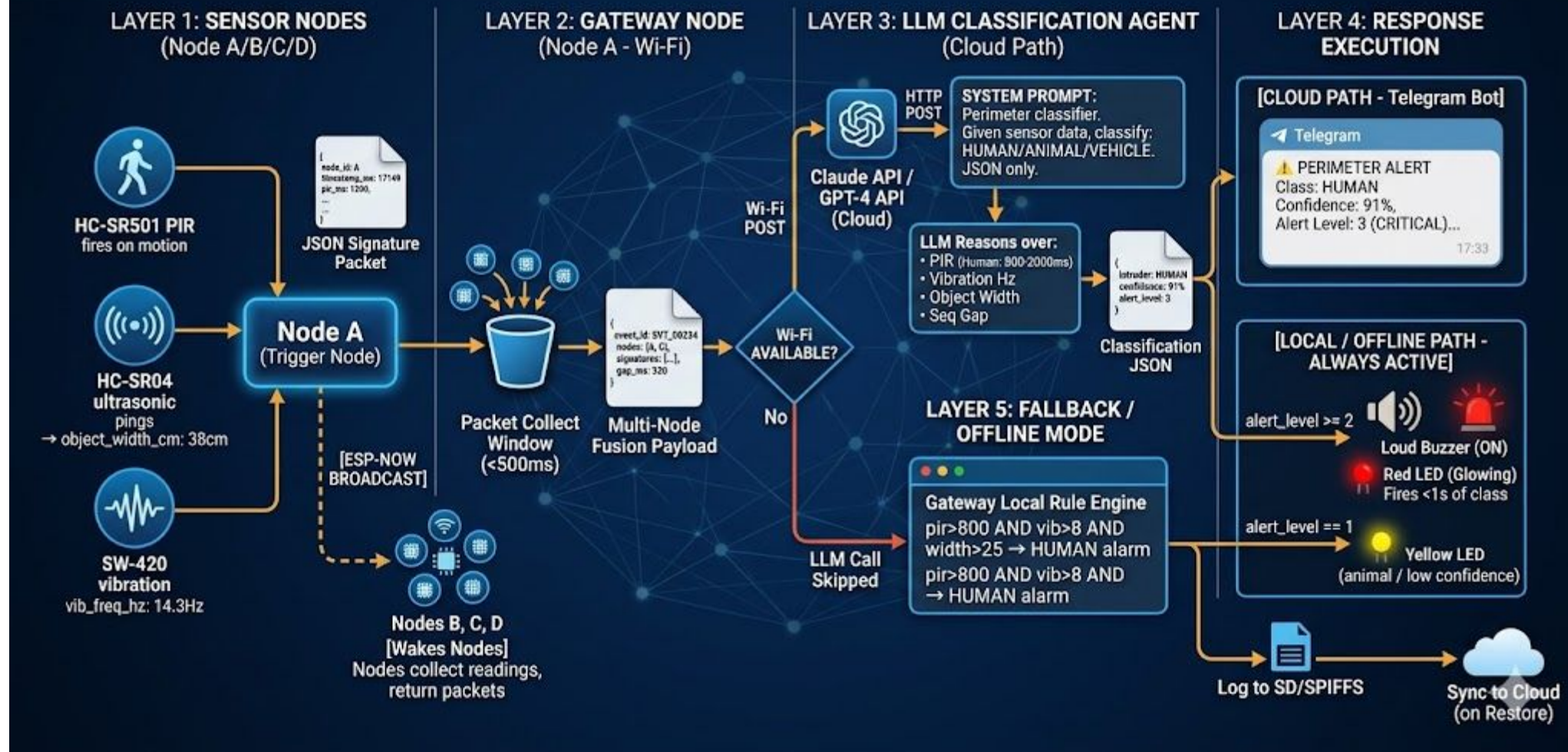
JSON payloads are analyzed by LLM agents (Anthropic/OpenAI) using signature-based prompts. Performance is measured via confusion matrices across 50 test events (human/animal/vehicle).

Tools and Resources

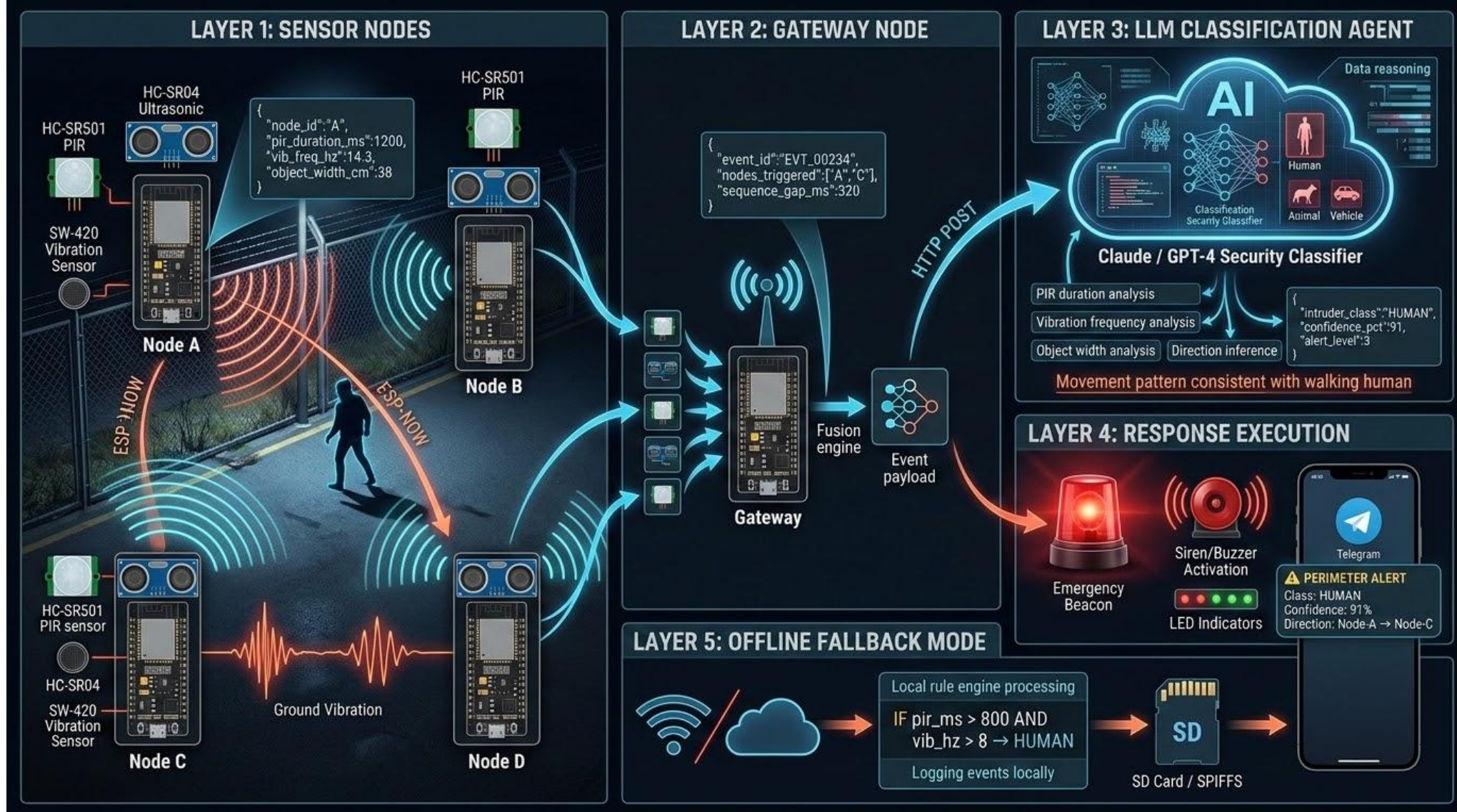
- Firmware: Arduino IDE / PlatformIO
- Bridge: Python Flask Gateway
- LLM: Claude API / GPT-4 API/Gemini/ Local Models
- Sensors: HC-SR501, SW-420, HC-SR04
- Alerts: Telegram Bot API

Methodology

SYSTEM ARCHITECTURE — EVENT FLOW



Methodology



Proposed Outcomes

Expected Results

- $\geq 85\%$ classification accuracy for human, animal, and vehicle intrusions.
- Alert latency below 4 seconds with ESP-NOW communication under 100ms.
- Offline buzzer activation within 1 second without internet dependency.
- System continues functioning even if one node fails.

Potential Impact

- Enables low-cost intelligent perimeter security using ESP32 hardware.
- Reduces false alarms through AI-based threat classification.
- Architecture can be adapted for flood detection, wildlife monitoring, and industrial safety systems.

Future Research

- On-device quantised ML models for offline classification.
- Large-perimeter scaling (8–12 nodes) with GPS tracking.
- ESP32-CAM integration for visual confirmation of threats.

Timeline of the Project

Initiation Phase (Weeks 1–2)

Define project goals, assign node responsibilities, procure components (ESP32, PIR, vibration, ultrasonic sensors), and conduct feasibility tests for ESP-NOW range.

Planning Phase (Weeks 3–4)

Develop firmware architecture, design JSON sensor payload schema, set up LLM API integration pipeline, and create a test event catalogue for classification.

Execution Phase (Weeks 5–10)

Flash firmware; validate ESP-NOW mesh; calibrate sensors (PIR, vibration, ultrasonic); integrate LLM classification agent; deploy Telegram Bot; conduct 50-event classification test with confusion matrix analysis; iterate on LLM prompt.

Closure Phase (Weeks 11–13)

Compile results, prepare IEEE paper draft, finalise demo setup, present at panel review, document firmware/API code on GitHub, and formally close the project.

Conclusion

Summary of Key Points

- Conventional PIR-based systems generate unclassified alerts, resulting in extremely high false alarm rates and delayed threat response.
- The proposed system uses a multi-node ESP32 ESP-NOW mesh with PIR, vibration, and ultrasonic sensor fusion for intelligent intrusion detection.
- An AI/LLM-based classification pipeline identifies intrusions as human, animal, or vehicle while supporting offline alarm generation and low-cost deployment.

Final Thoughts

The convergence of low-cost ESP32 hardware, the ESP-NOW protocol, and accessible LLM APIs makes this architecture feasible and timely. This project demonstrates that research-grade AI decision-making is no longer confined to high-end hardware — it is achievable on a ₹350 microcontroller with the right system design.

Next Steps / Road Map

- Complete a 3-node working prototype with labelled intrusion event testing.
- Develop an IEEE-format research paper and improve classification accuracy using larger datasets.
- Future upgrades include offline TinyML inference, GPS-based perimeter tracking, and ESP32-CAM visual verification.

References

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Thank You!